



Dual-TCR: A dual-branch target consistency reward framework for training-free personalized segmentation

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ABSTRACT

The Segment Anything Model (SAM) achieves strong zero-shot segmentation, but adapting it to recognize a user-specified object across images without manual prompting or retraining remains an open problem. Existing training-free methods extract a target embedding from a reference image and use feature similarity to generate automatic prompts. However, we find that both raw SAM and DINOv2 features produce unreliable localization in practice. The root cause is grid artifacts: periodic spatial distortions from patch tokenization that corrupt similarity maps and displace prompts onto background regions. To alleviate this, we introduce Dual-TCR, a framework that explicitly decouples semantic localization from geometric segmentation. Specifically, we construct spatially coherent similarity maps via NVIDIA RADIO to mitigate patch grid artifacts, providing reliable prompts that guide SAM purely as a geometric mask decoder. To autonomously resolve the ambiguity between competing mask candidates, we propose the Target Consistency Reward (TCR), a zero-shot arbitrator scoring each candidate via feature alignment with the reference target embedding. We further introduce a Hybrid Refining Module (HRM) step, fusing outputs from both branches to produce a refined final output mask. Evaluated on the PerSeg benchmark, Dual-TCR achieves 93.42% mIoU and 75.60% bIoU, establishing a new state-of-the-art specifically within the training-free category. Code is available at: <https://github.com/KaitoEight/dual-tcr>.

1. Introduction

Image segmentation has undergone a fundamental shift with the emergence of promptable foundation models [1]. The Segment Anything Model (SAM) [2] demonstrated that a single model trained on over one billion masks can segment any object when given an appropriate geometric prompt. Yet this promptability is also its primary limitation: SAM is semantically agnostic. It requires a human-provided prompt for each new image to specify the region of interest. Tasks such as isolating a specific pet across a vast photo album or tracking a custom tool in a robotics pipeline demand that the system *know* which object to segment based on its visual identity, rather than relying on explicit spatial coordinates.

Training-free personalization methods address this gap by transforming the segmentation paradigm into a visual matching problem. By extracting a target embedding from a single reference image, methods like PerSAM [3] utilize cosine similarity to automatically generate point prompts for new images. In theory, this single-shot retrieval approach is highly efficient; in practice, the quality of the dense feature embed-

ding and the spatial reliability of the resulting similarity map dictate the success of the entire pipeline.

We identify a systematic limitation in methods that rely on frozen vision backbones. Vision Transformers like DINOv2 [4] suffer from grid artifacts caused by discrete tokenization, which fragments spatial semantic continuity during dense matching. While the highest similarity response often pinpoints a valid target point, adjacent checkerboard fluctuations trigger unreliable negative prompts and degrade mask decoding, as shown in Fig. 1. Furthermore, while the native encoder of SAM produces geometrically aligned features, they lack the high-level semantic richness required to distinguish a target object from visually similar distractors across different environments. This limitation is well-documented: AM-RADIO [4] shows SAM features are 4× worse than DINOv2 on semantic retrieval and 1.7× worse on ADE20K linear probing, and SAMantics [5] independently confirms that SAM transfers poorly across a broad suite of semantic benchmarks.

To overcome the problem, we propose decoupling the two sub-tasks of personalization: semantic localization and geometric mask generation. NVIDIA RADIO [4], trained through multi-teacher distillation from

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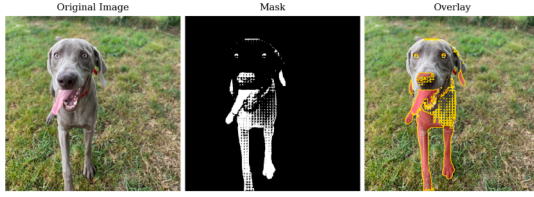


Fig. 1. Visualization of grid artifacts on the segmentation result: The mask is over-segmented into grid shapes due to the patch-based Vision Transformers (ViTs).

CLIP [1], DINOv2 [6], and SAM [2] with explicit spatial fidelity objectives, produces dense representations that are both semantically robust and free of grid artifacts. In our architecture, both SAM [2] and RADIO [4] act as parallel localizers that produce candidate point prompts. SAM [2] serves additionally as the shared geometric mask decoder. Because each branch yields a distinct prompt set, the framework generates two competing mask candidates whose ambiguity must be resolved automatically. Accordingly, we introduce the Dual Target Consistency Reward (Dual-TCR) framework for this purpose, which scores each candidate by measuring how closely the predicted region’s pooled features align with the reference target embedding in the feature space of SAM. To go beyond mere selection, we further apply a HRM that fuses high-confidence evidence from both branches, intersecting their masks, combining their point prompts, and blending their decoder logits weighted by TCR scores before invoking the SAM decoder a final time. The resulting mask is gated by a safety check that reverts to the winning candidate whenever fusion would degrade target consistency. The entire pipeline operates without any parameter updates or gradient descent.

The main contributions are summarized as follows. *First*, we propose a dual-branch framework with a target consistency reward mechanism for training-free personalized segmentation, dubbed Dual-TCR, to alleviate the oversegmentation problem of existing personalized segmentation frameworks. *Second*, we propose a zero-shot arbitrator scoring module (TCR) and a hybrid refining module (HRM) to summarize the final segmentation mask results better. *Third*, we provide an empirical analysis demonstrating that ViT grid artifacts are a primary cause of localization failure and that RADIO’s spatial coherence is paramount for zero-shot prompting. Evaluated on the PerSeg benchmark, our Dual-TCR achieves 93.42% mIoU and 75.60% bIoU, advancing the training-free personalized image segmentation.

2. Related work

2.1. Interactive image segmentation with prompt

Early segmentation models required dense pixel-level supervision [7] or interactive guidance through scribbles and bounding boxes [8]. These methods were highly effective within their training distributions but demanded per-image human effort, rendering them impractical for large-scale workflows. SAM [2] disrupted this paradigm by pre-training a ViT backbone alongside a lightweight prompt encoder on the SA-1B dataset [2], producing a promptable model that generalizes across domains without task-specific fine-tuning. While robust at geometric segmentation, SAM1 [2] requires external prompts and lacks semantic awareness. Rather than adopting the hierarchical backbones of SAM2 [9] or the multimodal design of SAM3 [10], our Dual-TCR builds on the dense spatial features of Plain ViT in SAM1. Our method assigns semantic localization to NVIDIA RADIO [4] and uses SAM1 purely as an accurate geometric mask decoder.

2.2. Personalized one-shot segmentation

Personalized segmentation adapts foundation models to recognize user-specified concepts. PerSAM [3] is the most direct predecessor to

our work, extracting a mean target embedding from the reference image and computing dense cosine similarity against test images to select positive and negative point prompts. Similarly, Matcher [11] integrates bidirectional feature matching across all-purpose vision models to guide promptable segmentation. Along this line, P²SAM [12] propagates part-level correspondences between the reference and target images to derive robust prompts for SAM. SegGPT [13] treats segmentation as an in-context visual learning problem, conditioning a generalist model on reference image-mask pairs. Painter [14] and VP [15] adapt in-context learning to visual tasks with modest results. While methods like SegGPT [13] achieve high accuracy (94.3% mIoU on PerSeg), they require a massive 354M-parameter backbone and a specific domain for training. Recent works adapt SAM to multimodal tasks via hybrid prompting or adapters, such as RGB-Thermal detection in HyPSAM [16] and KAN-SAM [17]. In contrast, Dual-TCR focuses on single-shot RGB personalization and operates strictly without training or parameter updates.

2.3. Ensemble and dual-branch architectures

Combining multiple predictions to improve robustness has a long history in computer vision. Classical ensemble methods aggregate outputs from independently trained models through voting or averaging [18]. More recent approaches, such as HyperSeg [19] and QueryInst [20], generate multi-scale proposals from different decoder heads and apply learned scoring functions to select or fuse the results. In contrast, our approach employs two structurally complementary branches: one optimized for geometric boundary cues and one for semantic localization. Arbitration between them requires no learned parameters, reusing only the frozen feature space already presented in the pipeline.

3. Proposed method

3.1. Problem formulation

Given a reference image $I_{ref} \in \mathbb{R}^{H \times W \times 3}$ and a binary mask $M_{ref} \in \{0, 1\}^{H \times W}$ indicating the target object, the goal of personalized segmentation is to identify and segment the same semantic concept in a target image $I_t \in \mathbb{R}^{H \times W \times 3}$. Formally, we aim to construct a mapping function $\mathcal{F} : (I_{ref}, M_{ref}, I_t) \mapsto M_t$, where $M_t \in \{0, 1\}^{H \times W}$ is the predicted target mask. This function must bridge visual disparities between the reference and target scenes, such as variations in scale, pose, illumination, and background clutter, using only the provided one-shot prompt.

Under the training-free paradigm, this mapping $\mathcal{F}$ operates without any iterative optimization. Let Θ be the parameters of the pre-trained foundation models. The entire pipeline is executed with strictly frozen weights ($\Delta\Theta = 0$ during inference). Consequently, without task-specific fine-tuning, the success of the segmentation relies entirely on the inherent representational capacity and spatial coherence of the pre-trained latent spaces.

3.2. Preliminaries: SAM and PerSAM

To lay the foundation for our approach, we briefly formulate the core mechanisms of SAM and the PerSAM baseline.

SAM Formulation. SAM [2] consists of a heavy image encoder Enc_I , a prompt encoder Enc_P , and a lightweight mask decoder Dec_M . For an image I and a set of spatial prompts P , the predicted mask M is generated via:

$$F_I = \text{Enc}_I(I), \quad F_P = \text{Enc}_P(P) \quad (1)$$

$$M = \text{Dec}_M(F_I, F_P) \quad (2)$$

where $F_I \in \mathbb{R}^{C \times h \times w}$ is the dense image feature, and $F_P \in \mathbb{R}^{N \times C}$ represents the sparse prompt embeddings with N denoting the number of input prompts.

PerSAM Formulation. The PerSAM [3] automates the generation of prompt P using visual matching. It extracts the reference feature map

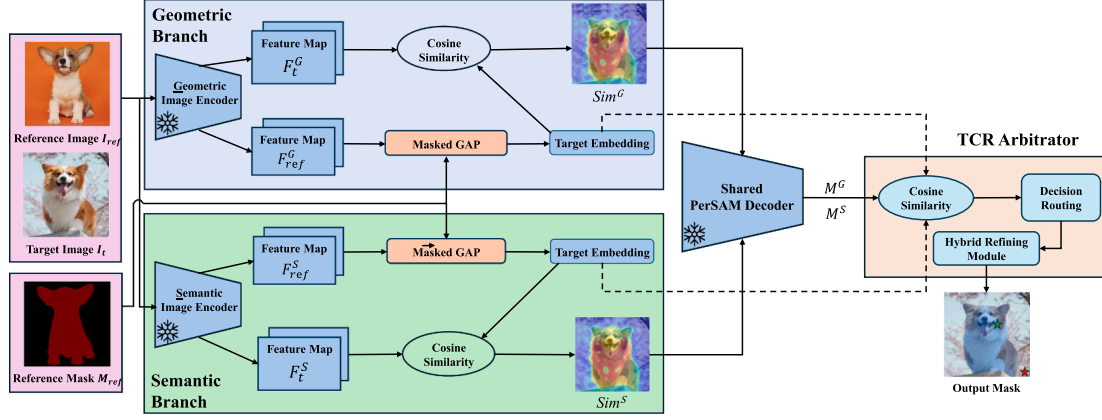


Fig. 2. Overview of our proposed Dual-TCR.

$F_{ref} = \text{Enc}_I(I_{ref})$. The target embedding $f_{ref} \in \mathbb{R}^C$ is computed by applying global average pooling (GAP) over the foreground region:

$$f_{ref} = \frac{\sum_{x,y} F_{ref}(x,y) \cdot M_{ref}(x,y)}{\sum_{x,y} M_{ref}(x,y)} \quad (3)$$

For a target image I_t , its feature map $F_t = \text{Enc}_I(I_t)$ is compared pixel-by-pixel against f_{ref} using cosine similarity, yielding a confidence heatmap $Sim \in \mathbb{R}^{1 \times h \times w}$:

$$Sim(x,y) = \frac{f_{ref} \cdot F_t(x,y)}{\|f_{ref}\| \|F_t(x,y)\|} \quad (4)$$

The spatial coordinates corresponding to $\max(Sim)$ and $\min(Sim)$ are extracted as the positive and negative prompts $P = \{p^+, p^-\}$, which are subsequently fed into the decoder of SAM. While mathematically elegant, this formulation assumes that F_t preserves the spatial continuity. As discussed, standard ViTs violate this assumption due to patch tokenization, motivating our proposed Dual-TCR.

3.3. The dual-TCR framework

As illustrated in Fig. 2, Dual-TCR abandons the single-encoder bottleneck. We execute two independent prompt generation pipelines in parallel, leveraging different foundation spaces, decode candidate masks through a shared PerSAM decoder, and resolve their ambiguity through TCR arbitration followed by Hybrid Refining Module (HRM). In this section, we use superscript G to denote quantities from the geometric branch (i.e., SAM encoder) and superscript S for the semantic branch (i.e., RADIO encoder).

3.3.1. Motivation

Our design shares the spirit of multi-candidate generation but differs in two fundamental ways. First, the two branches in Dual-TCR are structurally distinct rather than redundant: the geometric branch captures boundary cues aligned with the decoder's latent space, while the semantic branch captures spatially coherent semantic features free of grid artifacts. This complementarity is grounded in empirical evidence from AM-RADIO [4]: SAM features perform poorly in high-level object description, being 4× worse than DINOv2 [6] on semantic retrieval, whereas RADIO inherits and surpasses the semantic capabilities of both DINOv2 [6] and CLIP through multi-teacher distillation. Their outputs are therefore genuinely complementary; a mask that scores poorly from one perspective may score well from the other, which is precisely what the TCR arbitrator exploits. Second, all existing ensemble methods require a learned arbitrator or fusion module trained on labeled data. The TCR mechanism and HRM require no such supervision: they reuse the pre-trained SAM feature space already present in the pipeline, making the selection and fusion fully training-free. This combination of structural complementarity and zero-shot arbitration is, to our knowledge, not present in prior personalized segmentation work.

3.3.2. Dual-branch prompt processing

Geometric Branch. The primary pathway leverages the native ViT encoder of SAM, denoted as Enc^G . Given the reference image I_{ref} and its mask M_{ref} , we extract the dense feature map $F_{ref}^G \in \mathbb{R}^{C \times H^G \times W^G}$. The embedding $f_{ref}^G \in \mathbb{R}^{C^G}$ is obtained via GAP over the foreground region defined by M_{ref} :

$$f_{ref}^G = \text{GAP}(F_{ref}^G, M_{ref}) = \frac{\sum_{x,y} F_{ref}^G(x,y) \cdot M_{ref}(x,y)}{\sum_{x,y} M_{ref}(x,y)} \quad (5)$$

For a target image I_t , we compute the pixel-wise cosine similarity between f_{ref}^G and the target feature map F_t^G :

$$Sim^G(x,y) = \frac{f_{ref}^G \cdot F_t^G(x,y)}{\|f_{ref}^G\| \|F_t^G(x,y)\|} \quad (6)$$

yielding a dense heatmap Sim^G . The spatial coordinates corresponding to the absolute max/min similarity values are extracted as the positive p^{G+} and negative p^{G-} prompts, respectively. While these prompts inherently exist in the same latent space as the geometric decoder of SAM, ensuring strict structural compatibility, this branch is constrained by the class-agnostic pre-training of SAM. As documented by AM-RADIO [4] and SAMantics [5], SAM excels at edge detection but performs poorly on high-level semantic tasks, which is why Sim^G frequently suffers from weak semantic discriminability, particularly when distinguishing the target from visually analogous distractors in the background.

Semantic Branch. To overcome the semantic limitations of the geometric branch, the secondary pathway operates entirely within the feature space of NVIDIA RADIO [4], denoted as Enc^S . We independently extract semantic representations for both the reference and target images:

$$F_{ref}^S = \text{Enc}^S(I_{ref}), \quad F_t^S = \text{Enc}^S(I_t), \quad (7)$$

where $F_{ref}^S, F_t^S \in \mathbb{R}^{C^S \times H^S \times W^S}$. Following the identical pooling protocol, we derive the semantic target embedding $f_{ref}^S = \text{GAP}(F_{ref}^S, M_{ref})$ and construct its corresponding similarity map:

$$Sim^S(x,y) = \frac{f_{ref}^S \cdot F_t^S(x,y)}{\|f_{ref}^S\| \|F_t^S(x,y)\|} \quad (8)$$

Unlike standard self-supervised ViTs that exhibit severe patch-based discontinuities, the architecture of RADIO is optimized via multi-teacher knowledge distillation (incorporating CLIP [1] and DINOv2 [6]) with explicit high-resolution spatial objectives, surpassing both teachers on semantic benchmarks [4]. Notably, RADIOv2.5 [21] demonstrates that at 512×512 resolution, RADIO features most closely resemble those of DINOv2, expressing rich semantic content, while at higher resolutions they exhibit SAM-like sharp contour delineations, a resolution-adaptive

behavior ideally suited to our prompting stage. As a result, Sim^S maintains continuous spatial coherence and effectively mitigates grid artifacts. By executing an $\arg \max$ and $\arg \min$ over this spatially smoothed topological space, we obtain a secondary set of prompt coordinates, p^{S+} and p^{S-} .

3.3.3. Parallel mask generation

Once the spatial coordinates are extracted, both prompt sets are independently forwarded to the same frozen PerSAM decoder, which acts as a shared geometric backbone:

$$(M^G, L^G) = \text{Dec}^G(I_t, p^{G+}, p^{G-}), \quad (9)$$

$$(M^S, L^S) = \text{Dec}^G(I_t, p^{S+}, p^{S-}), \quad (10)$$

where L^G, L^S denote the predicted mask logits before binarization. Mask M^G originates from geometric cues, while M^S is guided by semantic localization of RADIO. Because the decoder is shared, the two candidates differ only in their spatial conditioning, which isolates the contribution of each encoder to the localization step.

3.3.4. TCR - target consistency reward arbitrator

The presence of two candidates necessitates a zero-shot decision mechanism. We introduce the Target Consistency Reward (TCR), which evaluates how well the predicted region of each candidate aligns with the reference target in the feature space of SAM. For each candidate mask $M \in \{M^G, M^S\}$, we project the mask onto the target feature space F_t^G and compute its GAP embedding:

$$f_t^G = \text{GAP}(F_t^G, M). \quad (11)$$

The TCR score for M is defined as the cosine similarity between $f_t^G(M)$ and the reference embedding f_{ref}^G :

$$\text{score}_M^G = \max \left(0, \frac{f_t^G(M) \cdot f_{ref}^G}{\|f_t^G(M)\| \|f_{ref}^G\|} \right). \quad (12)$$

Both candidates are scored by measuring feature alignment against the reference target. Rather than estimating an IoU, TCR operates as a relative binary arbitrator between two proposals synthesized by the SAM decoder. To provide a generalized scoring formulation across latent representations, we define the unified consistency reward parameterized by an interpolation weight $\gamma \in [0, 1]$:

$$s_M(\gamma) = (1 - \gamma) \cdot \text{score}_M^G + \gamma \cdot \text{score}_M^S, \quad (13)$$

where score_M^G and score_M^S denote the cosine similarity evaluated in the SAM geometric space and RADIO semantic space, respectively. By default, we adopt $\gamma = 0.0$ (SAM-space scoring) as the standard baseline configuration. Because the shared mask decoder operates natively in this geometric latent space.

Decision Routing. Let $w, \bar{w}$ refer to the winning, losing candidate selected by the TCR arbitrator, respectively. The arbitrator identifies the winning branch via:

$$w = \arg \max(\text{score}^G, \text{score}^S), \quad M_w \in \{M^G, M^S\}. \quad (14)$$

The higher-scoring candidate is the winner; its corresponding logits L_w and prompt set are propagated to the next stage.

3.3.5. HRM - hybrid refining module

A pure $\arg \max$ selection is robust but conservative: it discards information from the losing branch even when the two candidates agree on a high-confidence region. To exploit this latent agreement, we introduce a Hybrid Refining Module that fuses outputs from both branches and returns a single refined mask. The procedure consists of five sub-steps.

(a) Region Agreement. We define the high-confidence target region as the intersection of the two candidate masks:

$$f = M^G \cap M^S \quad (15)$$

Pixels in f are foreground under the similarity criteria of both encoders. Because both branches compute their similarity heatmaps based on target embeddings extracted from the ground-truth M_{ref} , this intersection represents a consensus constrained by target anchors.

(b) Combined Prompts. We assemble a richer prompt set by combining points from both branches. All positive points from the winner are kept; positive points from the losing part are kept only if they fall inside f ; all negative points from both branches are concatenated:

$$P^+ = \{p_w^+\} \cup \{p_{\bar{w}}^+ \mid p_{\bar{w}}^+ \in f\}, \quad (16)$$

$$P^- = \{p_w^-\} \cup \{p_{\bar{w}}^-\}. \quad (17)$$

This filtering keeps the contribution of the loser constrained to consensus regions while still leveraging its background-suppression cues.

(c) Weighted Logit Fusion. The decoder logits of the two candidates are blended in proportion to their TCR scores (computed by Eq. (12)):

$$L_{\text{hybrid}} = \lambda_w \cdot L_w + \lambda_{\bar{w}} \cdot L_{\bar{w}}, \quad \lambda_i = \frac{\text{score}_i}{\text{score}_w + \text{score}_{\bar{w}} + \epsilon}, \quad (18)$$

where $i \in \{w, \bar{w}\}$, $\epsilon = 10^{-6}$. To suppress the noise of the $\bar{w}$ candidate, we empirically set a threshold: if $\text{score}_{\bar{w}} < 0.3 \cdot \text{score}_w$, $L_{\text{hybrid}} = L_w$.

(d) Re-decode with Shared SAM. The combined prompts and fused logits are fed back into the SAM decoder, conditioned on the union bounding box of the two candidates whenever $\text{score}_{\bar{w}} \geq 0.5 \cdot \text{score}_w$, the refined output M_{hybrid} is formulated:

$$M_{\text{hybrid}} = \text{Dec}^G(I_t, P^+, P^-, \text{box} = \text{Bbox}(M^G \cup M^S), L_{\text{hybrid}}). \quad (19)$$

(e) Safety Rollback. M_{hybrid} is rescored using Eq. (12) if the score falls in the undefined range, the refinement is:

$$M^* = \begin{cases} M_{\text{hybrid}} & \text{if } \text{score}_{\text{hybrid}} \geq \text{score}_w - \tau \\ M_w & \text{otherwise} \end{cases} \quad (20)$$

We set $\tau = 0.02$ in our experiments; this prevents the fusion from degrading predictions when the two branches yield highly divergent mask candidates.

4. Experiments

4.1. Experimental setup

Dataset. We conduct comprehensive evaluations on the PerSeg benchmark [3]. The dataset covers 40 diverse object categories, spanning animals, household items, and architectural structures. Each category is represented by 5 to 7 high-resolution images capturing the object under extreme pose variations, occlusions, and diverse background contexts.

Evaluation Metrics. We evaluate the segmentation results using mean Intersection-over-Union (mIoU), boundary Intersection-over-Union (bIoU) [23], and mean Pixel Accuracy (mAcc). Unlike standard mIoU, bIoU is highly sensitive to boundary adherence and prompt placement errors.

Implementation Details. The semantic branch utilizes the ViT-L/16 variant of NVIDIA RADIO [4], while the geometric branch is built upon the ViT-H backbone of SAM [2]. Target and reference images are resized to 512×512 with standard ImageNet normalization during semantic feature extraction. For the DINOv2 ablation baseline, inputs are resized to 518×518 to maintain 14×14 patch divisibility. For mask decoding, inputs are padded to 1024×1024 resolution of SAM. The Hybrid Refining stage uses a logit-fusion gate of $0.3 \cdot \text{score}_w$, a union-box gate of $0.5 \cdot \text{score}_w$, and a rollback margin $\tau = 0.02$. All experiments were executed on a single NVIDIA RTX A4000 GPU (16GB VRAM) with zero training or gradient optimization.

4.2. Comparison with state-of-the-art methods

Table 1 reports the quantitative performance of Dual-TCR against recent methods. To ensure a fair evaluation, Table 1 categorizes methods

Table 1

Comparison with state-of-the-art personalized segmentation methods on the PerSeg benchmark. Baselines are explicitly categorized into fully supervised, parameter-efficient, and training-free tiers.

Category	Method	Year	mIoU	bIoU	Params.	#Pos.Prompts	Can	Barn	Clock	Cat	Backpack
Fully Supervised	VP [15]	2022	65.90	25.50	383M	0	61.2	58.6	59.2	76.6	66.7
	Painter [14]	2023	56.40	42.00	354M	0	19.1	3.2	42.9	94.1	88.1
	SEEM [22]	2023	87.10	55.70	341M	0	65.4	82.5	72.4	91.1	94.1
	SegGPT [13]	2023	94.30	76.50	354M	0	96.6	63.8	92.6	94.1	94.4
Param-Efficient	PerSAM-F [3]	2024	95.33	78.20	2	1	96.7	97.5	96.1	92.3	95.5
Training-Free	PerSAM [3]	2024	89.30	71.70	0	1	96.2	38.9	96.2	90.7	95.4
	Matcher [11]	2024	90.50	72.30	0	8	–	–	–	–	–
	P ² SAM [12]	2025	95.70	78.60	0	5	–	–	–	–	–
	Dual-TCR (Ours)	2026	93.42	75.60	0	1	97.2	96.2	93.4	95.6	96.3

Table 2

Ablation study on different encoder configurations compared to our proposed Dual-TCR. The compared components are single branch, and without HRM.

Configuration	Encoder			mIoU(%)
	SAM	RADIO	DINOv2	
Geometric Branch	✓			89.30
Semantic Branch		✓		89.16
			✓	85.83
Dual-TCR (Ours)	✓	✓		89.62

Table 3

Quantitative 1-shot segmentation comparison on the FSS-1000 benchmark following the standard split [25]. Baselines are divided by supervision level to ensure fair evaluation. All training-free models operate zero-shot without fine-tuning.

Category	Method	#Pos. Prompts	Training	mIoU (%) ↑
Fully Supervised	PANet [24]	0	✓	72.80
	SegGPT [13]	0	✓	86.10
Training-Free	PerSAM [3]	1	×	71.20
	Matcher [11]	8	×	82.10
	P ² SAM [12]	5	×	82.40
	Dual-TCR (Ours)	1	×	76.95

Table 4

Ablation study on HRM components. All variants share the same dual-encoder backbones and with TCR.

Dual-branch	TCR	HRM		mIoU (%)	bIoU (%)
		w/ Fusion	w/ Safety rollback		
✓	✓	–	–	91.40	73.20
✓	✓	✓	–	92.45	74.30
✓	✓	✓	✓	93.42	75.60

into fully supervised, parameter-efficient, and training-free tiers. Among training-free approaches, Dual-TCR outperforms the PerSAM baseline, improving mIoU by 4.12% and bIoU by 3.90% without introducing a single trainable parameter and utilizing only a pair of 1 positive and 1 negative prompts. Meanwhile, Matcher [11] uses up to 8 prompts, and P²SAM [12] uses up to 5 prompts, raising the number of input guidance compared to our work. In comparison with other fully supervised methods, such models required further domain-specific training on more than 300M parameters, including SegGPT [13].

To generalize the claim beyond PerSeg, Table 3 reports 1-shot performance on the large-scale FSS-1000 [25] benchmark following the standard evaluation protocol. The baselines are divided by supervision level to ensure a fair comparison. All training-free models operate zero-shot without fine-tuning. While supervised methods, SegGPT [13] achieves higher mIoU through extensive training, our Dual-TCR establishes a

strong training-free improvement over the baseline PerSAM [3]. Our Dual-TCR attains 76.95% mIoU, improved over the reproduced PerSAM baseline (71.20%) by +5.75% mIoU. The compared Matcher [11] and P²SAM [12] use the publicly reported results from the original papers.

4.3. Ablation studies

4.3.1. Impact of semantic encoders

To validate the hypothesis regarding grid artifacts, we isolate the localization encoder by swapping it within a PerSAM pipeline (single branch, without TCR or HRM). As shown in Table 2, substituting the native SAM encoder with DINOv2 [6] drastically drops mIoU to 85.83%. Despite the superior global semantic understanding of DINOv2, its patch tokenization artifacts severely disrupt dense pixel matching. This is consistent with AM-RADIO [4] and SAMantics [5], which independently confirm that SAM features perform poorly on semantic tasks, and no single encoder satisfies both localization and semantic discriminability simultaneously. Conversely, deploying RADIO as the sole semantic branch encoder matches the geometric branch baseline almost perfectly (89.16% vs. 89.30%), confirming that its spatial coherence is comparable to the native geometric alignment of SAM while inheriting far stronger semantic capabilities [4,21]. The leap to 93.42% mIoU only materializes when the geometric and semantic branches operate in parallel under TCR arbitration with HRM, demonstrating that a secondary candidate combined with an autonomous referee and a fusion step is the key to escaping the bottleneck of training-free personalization.

To quantitatively assess grid artifact impacts, Table 5 reports prompt hit rate, distance to mask center, spatial coherence, and gradient magnitude. While DINOv2 [6] yields the highest point-level hit rate (67.13%), its low spatial coherence (0.5465) reflects severe patch fragmentation. In contrast, SAM’s prompt localization is the lowest (58.33% hit rate) but still has the ability to generate a high-quality boundary mask, combined with RADIO [4], which achieves superior spatial coherence (0.8774), yielding clean topological blobs essential for stable mask decoding.

4.3.2. Ablation study on HRM contributions

Table 4 isolates the contribution of each component within the HRM module on top of the dual-branch baseline with TCR arbitration (91.40% mIoU). Integrating weighted logit fusion improves the performance to 92.45% mIoU, indicating that fusing complementary boundary cues from the loser mask refines the overall segmentation. Furthermore, enabling the target-consistency safety rollback yields an additional +0.97% mIoU, achieving 93.42% mIoU and 75.60% bIoU in the full pipeline.

4.3.3. Ablation study on TCR arbitration latent spaces

We investigate the choice of scoring latent space in Table 6 via the interpolation parameter $\gamma \in [0, 1]$. Setting $\gamma = 1.0$ (RADIO-space scoring) achieves the peak performance of 93.42% mIoU and 75.60% bIoU, leveraging rich distilled semantic representations to resolve subtle target-distractor ambiguities. Meanwhile, our compute-efficient baseline of

Table 5

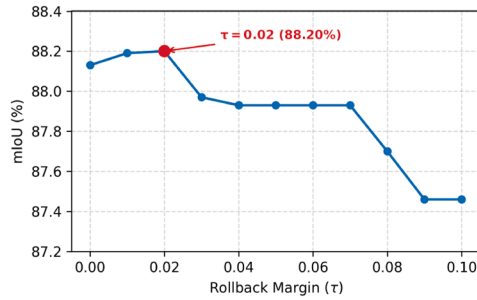
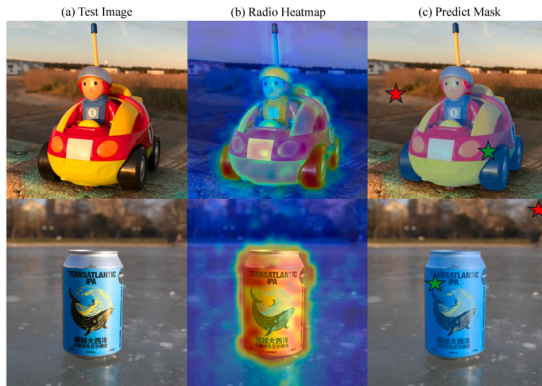
Quantitative comparison of different encoders on localization accuracy and spatial map quality on PerSeg [3].

Encoder	Prompt Hit Rate (%) $\uparrow$	Normalized Distance $\downarrow$	Spatial Coherence $\uparrow$	Gradient Magnitude $\downarrow$
SAM (ViT-H)	58.33	0.2640	0.7183	0.1393
DINOv2 (ViT-L)	67.13	0.2477	0.5465	0.1275
RADIO (ViT-L)	62.96	0.2575	0.8774	0.1605

Table 6

Ablation study on TCR arbitration latent spaces.

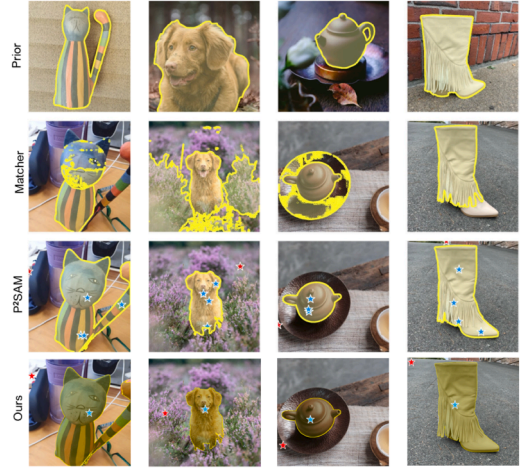
Scoring Space	Weight (γ)	mIoU (%)	bIoU (%)
SAM-space	0.0	92.34	74.18
DUAL-space	0.5	92.46	74.30
RADIO-space	1.0	93.42	75.60

**Fig. 3.** Sensitivity analysis of the HRM rollback margin τ .**Fig. 4.** Qualitative results of Dual-TCR. From left to right: Target image, RADIO heatmap, and predicted mask. Green and red stars denote positive and negative prompts, respectively.

$\gamma = 0.0$ maintains a highly competitive 92.34% mIoU ($< 1\%$ margin), confirming that scoring in the native decoder representation introduces no systematic bias while eliminating additional semantic pooling latency during deployment.

4.3.4. Sensitivity of the rollback margin τ

We evaluate the sensitivity of HRM to the rollback margin $\tau \in [0.00, 0.10]$, with results summarized in Fig. 3. Setting $\tau = 0.00$ enforces strict gating, which yields 88.13% mIoU in isolated ablation settings. This proves overly conservative, as it discards borderline-equivalent fusions that may offer better local geometries. The optimal balance is achieved at $\tau = 0.02$ (88.20%). Conversely, permissive gating ($\tau \geq 0.03$) leads to progressive performance degradation (dropping to 87.46%), confirming that our empirical choice of $\tau = 0.02$ robustly balances the exploration of consensus features while safely guarding against structural degradation.

**Fig. 5.** Qualitative comparison between Dual-TCR and state-of-the-art methods on the PerSeg benchmark. Blue and red stars denote positive and negative prompts, respectively.

4.4. Qualitative analysis

Fig. 4 demonstrates how each stage resolves distinct challenges: the RADIO semantic heatmap suppresses diffuse activations to locate cluttered targets (everage can), while the TCR arbitrator and HRM overcome part-bias on complex silhouettes (toy car) to produce complete, boundary-accurate masks. Fig. 5 compares Dual-TCR with competitive training-free baselines Matcher [11], and P²SAM [12]. While Matcher captures background distractors and P²SAM produces competitive segmentations, our Dual-TCR achieves finer boundary adherence and superior target coherence across challenging poses.

5. Discussion and limitations

Despite its robustness, our Dual-TCR introduces a computational trade-off, with the simultaneous execution of the SAM ViT-H geometric branch and RADIO ViT-L semantic branch nearly doubling feature extraction time to approximately 1.2 s per image on an NVIDIA RTX A4000 GPU. While the HRM stage remains lightweight (< 0.1 s), this overhead is justified for high-precision tasks that preclude domain-specific fine-tuning. However, the framework can underperform single-branch baselines in scenarios with extreme scale variation or repetitive thin structures. Addressing these failures via multi-scale prompting or feature selection remains an avenue for future research.

6. Conclusion

In this work, we introduced Dual-TCR, a training-free, dual-branch framework redefining personalized image segmentation by decoupling semantic localization from geometric mask generation. By assigning semantic localization to the RADIO semantic branch and geometric decoding to the shared SAM backbone, our approach addresses the fundamental architectural tension inherent in single-branch baselines. The Target Consistency Reward and Hybrid Refining Module promote robust arbitration and consensus mask fusion, resulting in significant performance

gains across diverse object types without requiring any parameter updates. Our Dual-TCR sets a new benchmark for personalized segmentation. Future work focuses on distilling this dual-branch synergy into a unified backbone.

CRedit authorship contribution statement

Minh-Khanh Le: Writing – original draft, Visualization, Validation, Software, Investigation; **Thanh-Danh Nguyen:** Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization; **Vinh-Tiep Nguyen:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization; **Tam V. Nguyen:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data is available with this publication: R. Zhang, Z. Jiang, Z. Guo, S. Yan, J. Pan, X. Ma, H. Dong, P. Gao, H. Li, Personalize segment anything model with one shot, ICLR (2024).

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